BEYOND LOCAL LYAPUNOV STABILITY: INVERSE MOMENTS AND NONLINEAR RETURN IN SGD

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ABSTRACT

Stochastic gradient descent can remain confined while local linear models predict expansion. We separate the return mechanism creating a stationary distribution from the moments describing its central shape. Scalar quadratic SGD gives $a_B = 1 - \eta h_B$: positive moments govern perturbation growth, while a negative root $\mathbb{E}|a_B|^{-\alpha} = 1$ describes small stationary amplitudes in a returned process. The harmonic reference $\mathbb{E}|a_B|^{-1} = 1$ separates central concentration from depletion under explicit assumptions. A matrix renewal formulation retains direction dynamics. Coercive nonconvex SGD constructions then establish attracting invariant laws beyond local expansion, including a control with a positive stationary tangent exponent. Neural batch-size–step-size heatmaps and fixed-frame continuations show distinct scalar regimes and alternating two-lobe occupancy while transported tangents grow and loss decreases. These results identify what can coexist beyond a local edge and why neural bimodality needs direct, time-resolved measurement.

1 INTRODUCTION

A neural network can keep learning while local curvature predicts expansion. Can the optimizer remain statistically confined, and what shape does its occupation distribution have? An expanding deterministic quadratic has no return mechanism. Random curvature further separates the boundaries for mean-square, logarithmic, and inverse-moment growth.

A scalar quadratic gives the multiplier $a_B = 1 - \eta h_B$, with sampled curvature h_B and step size η. Its moments $m(p) = \mathbb{E}|a_B|^p$ separate several questions. Positive orders describe perturbation growth; the logarithmic average describes a typical product. The negative moment $m(-1)$ measures concentration toward zero. We call $H = 1/m(-1)$ the *harmonic gain*; its central-shape interpretation requires an explicit return model.

Classical Kesten–Goldie theory describes large values of a contractive random affine recursion. Our central-mass formulation combines local expansion with outer return: a *negative* moment root then controls small stationary amplitudes. We use reflected-walk and renewal ideas, retain the evolving direction in a matrix formulation, and exhibit nonconvex SGD losses whose gradients supply return.

We separate local growth, nonlinear recurrence, and coordinate-distribution shape through three results:

1. A scalar-to-matrix central renewal formulation with explicit return assumptions. It identifies the inverse root and explains when the harmonic reference separates central concentration from depletion.
2. Concrete coercive-loss SGD constructions proving that nonlinear return permits an attracting invariant law beyond local expansion. A separate piecewise-smooth construction also has a provably positive stationary tangent exponent.
3. Neural heatmaps and fixed-frame continuations revealing a concrete diagnostic disagreement: an MLP has negative late fixed-axis log gain, positive transported growth, and alternating two-lobe occupancy under iid batches. A short CNN candidate fails confirmation.

The constructions prove what can coexist; the neural measurements test these distinctions over finite training windows.

2 SCALAR QUADRATICS AND THE NEGATIVE-MOMENT BOUNDARY

2.1 FROM ONE DETERMINISTIC EDGE TO SEVERAL STOCHASTIC DIAGNOSTICS

For deterministic $L(x) = hx^2/2$ with $h > 0$, the entire orbit is

$$x_t = (1 - \eta h)^t x_0. \tag{1}$$

The condition $0 < \eta h < 2$ contracts every nonzero start to zero. At $\eta h = 2$ the sign alternates without decay; above two the magnitude diverges. A fixed quadratic therefore cannot explain bounded large-step fluctuations after its expanding boundary. Random curvature changes the product law, and nonlinear curvature can change the orbit's return.

On the scalar sampled quadratic $\ell_B(x) = h_B x^2/2 - \xi_B x$, SGD is exactly

$$X_{t+1} = A_{t+1} X_t + \eta \xi_{t+1}, \qquad A_B = 1 - \eta h_B. \tag{2}$$

Two trajectories using the same sampled batches differ by $\Delta_{t+1} = A_{t+1}\Delta_t$. Write $M = |A_B|$, so moments always use the absolute multiplier, and define

$$m(p) = \mathbb{E} M^p, \quad \gamma = \mathbb{E} \log M, \quad H = \frac{1}{m(-1)}. \tag{3}$$

Here H is the harmonic gain; its inverse $m(-1)$ is the negative-moment coefficient. We only assign a finite inverse-moment estimate this meaning when the population inverse moment exists. For constant curvature and $p \neq 0$, $m(p)^{1/p} = M$, $\gamma = \log M$, and $H = M$: the boundaries all coincide at $|1 - \eta h| = 1$. Curvature randomness separates them.

Proposition 1 (Scalar moment ordering). *Let $M > 0$, $\mathbb{E}|\log M| < \infty$, and suppose the displayed moments are finite. Then*

$$\log H \ \le\ \gamma \ \le\ \log m(1) \ \le\ \tfrac{1}{2} \log m(2). \tag{4}$$

For iid multipliers and $\Delta_0 \neq 0$, $t^{-1} \log |\Delta_t/\Delta_0| \to \gamma$ almost surely and $\mathbb{E}|\Delta_t/\Delta_0|^p = m(p)^t$. Consequently $H > 1$ implies $\gamma > 0$. The converse need not hold.

The use of absolute values matters: $m(1)$ is a mean-absolute multiplier, not the signed mean $\mathbb{E} a_B$. For $p > 0$, $m(p) < 1$ means contraction of a pth perturbation moment. For $p < 0$ it means contraction of an *inverse distance*, so it measures repulsion from zero, rather than contraction toward zero. The logarithmic criterion lies between these orders. We use the term harmonic expansion for $H > 1$ while keeping this direction of the inequality explicit.

A concrete two-curvature law already separates the references. Let $a_B = -1/2$ with probability u and $a_B = -2$ otherwise; both are realizable with positive quadratic curvatures. Then

$$\gamma = (1 - 2u) \log 2, \qquad H = (1/2 + 3u/2)^{-1}. \tag{5}$$

For $1/3 < u < 1/2$, the product is logarithmically expanding but $H < 1$. For $u < 1/3$, both diagnostics expand. No dimension or gradient-alignment assumption is needed for this separation.

The classical Kesten–Goldie result concerns the *affine* recursion (2): with $\gamma < 0$, nondegenerate additive forcing, and the usual renewal assumptions, its stationary upper tail has exponent $\kappa > 0$ solving $m(\kappa) = 1$ (Kesten, 1973; Goldie, 1991). This is a baseline, not a theorem of post-expansion confinement. In particular, the nondegenerate Gaussian affine model used in Appendix A has no invariant probability when $\gamma > 0$. An additional return mechanism is needed.

2.2 NONLINEAR RETURN CHANGES AN UPPER-TAIL QUESTION INTO A CENTRAL-MASS QUESTION

An *invariant law* π satisfies $X_0 \sim \pi \Rightarrow X_t \sim \pi$ at every time. It is *attracting* when other initial laws converge to it in distribution. These statements concern the law of a state; they need not imply contraction of paths using the same random updates.

The simplest exact return model is saturated amplitude:

$$R_{t+1} = \min\{r_*, M_{t+1} R_t\}, \qquad 0 < R_t \le r_*. \tag{6}$$

Here $r_* > 0$ is the outer return radius. The model separates local multiplicative motion from return at this scale. It is an explicit model, not an identity for a globally quadratic loss.

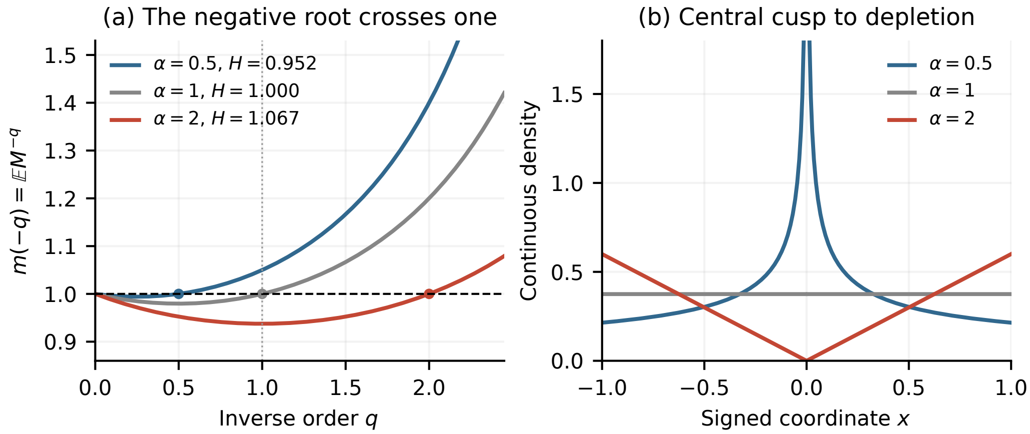

Figure 1: **An exact central-shape law for a returned multiplier.** Let $M = e^{V-U}$ with independent $U \sim \mathrm{Exp}(b)$, $V \sim \mathrm{Exp}(3)$, and $b = 3 + \alpha$. The saturated process $R^+ = \min\{1, MR\}$ has positive local log drift in all three cases. (a) Its inverse root $m(-\alpha) = 1$ crosses order one as H crosses one. (b) The continuous part of the symmetrically signed invariant law is $(3/b)(\alpha/2)|x|^{\alpha-1}$ on $0 < |x| < 1$: a cusp, a flat center, or central depletion. Each endpoint also has an atom of mass $(1 - 3/b)/2$, which is not drawn as a density. These are analytic curves; Proposition 7 in Appendix A verifies the law exactly.

Theorem 2 (An inverse-moment law after local expansion)**.** *Suppose* $\mathbb{E}|\log M| < \infty$*,* $\gamma > 0$*,* $\log M$ *is nonarithmetic, and* $m(-\alpha) = 1$ *for some* $\alpha > 0$*. Assume* $m(-q)$ *is finite in a neighborhood of* α *and* $0 < -\mathbb{E}[M^{-\alpha} \log M] < \infty$*. Then (6) has a unique invariant probability on* $(0, r_*]$*, and for some* $C \in (0, \infty)$*,*

$$\mathbb{P}_\pi(R \le r) \sim C(r/r_*)^\alpha, \qquad r \downarrow 0. \tag{7}$$

If, additionally, $\log M$ *has a bounded continuous density of compact support, then* $p_R(r) \sim C\alpha r_*^{-\alpha} r^{\alpha-1}$*. When the negative-moment curve is strictly convex and finite through* 1 *and* α*,*

$$H < 1 \iff \alpha < 1, \qquad H = 1 \iff \alpha = 1, \qquad H > 1 \iff \alpha > 1. \tag{8}$$

Thus positive local log growth can coexist with a central density cusp ($0 < \alpha < 1$); the harmonic crossing separates this cusp from central depletion ($\alpha > 1$) in this return model. For a reflection-symmetric signed coordinate, the same power occurs on each side. Central depletion is a local shape conclusion. Exactly two modes additionally requires control of the outer density, and arbitrary reinjection can reverse the conclusion (Appendix A.7).

Proof idea. The change of variable $W = \log(r_*/R)$ gives $W_{t+1} = \max\{0, W_t - \log M_{t+1}\}$. This is a reflected random walk with negative mean increment. Its invariant law is the supremum of the unreflected walk. The nonarithmetic assumption rules out log-radius lattice effects. Tilting the increment law by $M^{-\alpha}$ reverses its drift; the overshoot renewal theorem gives an exponential tail in W, hence the power (7) in R. The root therefore has the opposite sign from the Kesten upper-tail root.

3 A MATRIX RENEWAL FORMULATION

3.1 RANDOM MATRICES: RADIUS AND DIRECTION MUST EVOLVE TOGETHER

For a quadratic in d parameters, the linear update is $U_{t+1} = A_{t+1} U_t$, where $A = I - \eta H_B$. Writing $U = rs$, with $\|s\| = 1$, gives

$$r^+ = r\|As\|, \qquad s^+ = As/\|As\|. \tag{9}$$

Thus a matrix changes both the amplitude and the direction of the next update. Its log gains need not be independent. The appropriate analogue of the scalar moment $\mathbb{E}|a|^q$ is the growth rate of the positive operator

$$(P_q f)(s) = \mathbb{E}\left[\|As\|^q f\left(\frac{As}{\|As\|}\right)\right], \qquad k(q) = \mathrm{spr}(P_q). \tag{10}$$

Here spr denotes spectral radius: $k(q)$ is the long-run moment-growth factor, including changes of direction. The operator weights each next-direction value of f by the current radial gain to order q. When the original angular chain is ergodic, identifies the top product exponent, and the pressure is differentiable at zero, $\gamma = (\log k)'(0)$ is the linear-product Lyapunov exponent. In one dimension, $k(q) = \mathbb{E}|a|^q$. Classical matrix renewal theory uses a positive root to describe large values of a contractive affine recursion (Kesten, 1973; Guivarc'h & Le Page, 2016). The following result uses a negative root to describe small amplitudes of a locally expanding process supplied with return. Its proof changes the sampling law (a probability *tilt*) to emphasize rare inward excursions. The assumptions ensure that directions mix and that this tilted radius drifts inward; Appendix B gives the formulas and an explicit noncommuting example.

Theorem 3 (Matrix central law with finite return cycles)**.** *Suppose the linear product is locally expanding, $\gamma > 0$, and $k(-\alpha) = 1$ for some $\alpha > 0$. Assume the negative-order spectral and Markov renewal conditions in Assumption 1, including a bounded strictly positive eigenfunction and nonarithmetic angular mixing. Start an exact linear excursion below radius r_0 and restart it from the same independent entry law on its first exit above r_0. If the mean excursion length is finite and the entry law has a finite inverse α-moment, then its invariant law satisfies*

$$\lim_{r\downarrow 0} \frac{\pi(\|U\| \le r,\ U/\|U\| \in D)}{(r/r_0)^\alpha} = C\chi(D), \tag{11}$$

for angular continuity sets D, provided the tilted excursion has a positive probability of never returning to the upper boundary. Here $C > 0$ and χ is a probability measure on directions. The matrices may be noncommuting.

What this establishes. Finite return cycles create stationarity; the negative moment root determines how often stationary excursions reach the center. The proof tilts the joint radius–direction process, applies a Markov renewal theorem, and subtracts occupation after upper exit. It does not replace matrix products by Rayleigh quotients. For a genuine nonlinear SGD chain, Proposition 9 gives a precise transfer criterion in terms of its entry law and local transition kernel. Those conditions are additional to positive local growth and are not established for our neural networks.

Which density can lose its central peak? With an additional renewal-density regularity assumption, (11) gives radial density $p_R(r) \asymp r^{\alpha-1}$. If a joint local density limit in radius and direction also holds, dividing by the polar volume factor gives ambient density $p_U(rs) \asymp r^{\alpha-d}$ along directions with positive limiting density. For a strictly convex log pressure with roots $-\alpha$ and 0, $k(-1) < 1$ means $\alpha > 1$, whereas $k(-d) < 1$ means $\alpha > d$. A fixed coordinate integrates over directions and has no such universal threshold: Proposition 10 gives a stationary nonlinear SGD example with both inequalities but centrally unimodal fixed-coordinate marginals. This distinction motivates direct fixed-direction measurements in the neural experiments.

4 Nonconvex SGD supplies its own return

The next construction makes return part of the sampled loss. Here the local exponent freezes a Jacobian at the population minimizer; the occupied exponent follows Jacobians along a stationary trajectory.

To prove that return can arise from a gradient update, rather than from an imposed cap, draw $h \in \{.95, 1.05\}$ equiprobably and $e \sim N(0, \sigma^2)$, independently, with $\sigma > 0$. Use

$$\ell(x; h, e) = \tfrac{1}{2}(\sqrt{h}\,x - e)^2 - \tfrac{.92}{2}\{x^2 - \log(1 + x^2)\}. \tag{12}$$

Every sampled loss is coercive: it tends to infinity as $|x| \to \infty$. The population loss, up to a constant, is $L(x) = .04x^2 + .46\log(1 + x^2)$, with unique minimizer zero but negative curvature on part of the line.

Theorem 4 (Smooth nonconvex SGD can return despite central expansion)**.** *More generally, replace $.92$ in ([12]) by β and draw $h = \beta + \mu \pm s$ equiprobably, with $0 < s < \mu$ and $\beta > 8\mu$. At batch size one, every step $0 < \eta < 2/(\mu + s)$ has a unique attracting invariant probability with a smooth, strictly positive, even density and all finite polynomial moments. Its central Jacobian is $1 - \eta h$, whereas its outer slopes are $1 - \eta(\mu \pm s)$.*

For $(\beta, \mu, s, \eta) = (.92, .08, .05, 2.015)$, the central absolute gains are $.91425, 1.11575$, and

$$\gamma_0 = 0.009938\ldots > 0, \qquad H_0 = 1.004999\ldots > 1, \qquad \alpha_0 = 2.017427\ldots > 1, \tag{13}$$

where $\frac{1}{2}(.91425^{-\alpha_0} + 1.11575^{-\alpha_0}) = 1$. The outer slopes are $.93955, .73805$, both strictly contracting.

Proof idea. The exact update is $X^+ = A_\infty X - \eta\beta X/(1+X^2) + \eta\sqrt{h}\, e$, with $A_\infty = 1 - \eta(h-\beta)$. The middle term is bounded and the outer slopes contract. For every $p > 0$, the bound $\mathbb{E}[|X^+|^p \mid X = x] \le c_p|x|^p + C_p$, with $c_p < 1$, controls excursions to infinity. This inward drift and the positive Gaussian transition density give recurrence and uniqueness. At the displayed step, the local random product has positive mean log gain because curvature is much larger near the origin. This theorem proves recurrence after the *central* Lyapunov and harmonic boundaries; it does not identify the tangent exponent along the invariant trajectory or infer its modes. These are separate empirical measurements below.

For this example α_0 describes the *frozen* multiplier law. At every fixed $\sigma > 0$, the actual invariant density is strictly positive at zero, so a zero-noise power-law depletion cannot be inferred literally as $x \to 0$. The recurrence theorem and the central renewal theorem establish complementary statements, with different hypotheses.

4.1 What balances at equilibrium?

For plain SGD $X^+ = X - \eta g_B(X)$, stationarity with finite second moments gives the exact identity

$$2\mathbb{E}_\pi\langle X, \nabla L(X)\rangle = \eta \mathbb{E}_{\pi,B}\|g_B(X)\|^2. \tag{14}$$

Indeed, expand $\|X^+\|^2 - \|X\|^2$, condition on X, and integrate against π. The identity is justified for Theorem 4, whose loss gradients have linear growth and whose invariant law has all moments. Inward motion on the left balances the squared random updates on the right. Neither side is determined by the Jacobian at zero. The equilibrium is a distribution over visited states, rather than a point where the frozen multiplier must equal one.

The balance is visible directly in the conditional radial drift of the smooth example. At batch one, define $d(x) = \mathbb{E}[(X^+)^2 - X^2 \mid X = x]$. Independence and the centered residual give

$$d(x) = \left[\left(1 - \eta\mu - \frac{\eta\beta}{1+x^2}\right)^2 + \eta^2 s^2 - 1\right] x^2 + \eta^2\sigma^2(\beta + \mu). \tag{15}$$

For the displayed parameters, small states have outward drift, whereas $d(x)$ is negative at sufficiently large $|x|$. The invariant law averages these regions so that $\mathbb{E}_\pi d(X) = 0$. A zero of $d(x)$ is a local balance radius, not an assertion that the distribution is a point mass there.

In the smooth construction, the outer slopes control return while the central sampled curvatures control frozen expansion. A separate exact control shows that the *occupied* tangent exponent can be positive as well. Let $a \in \{.58, .62\}$ equiprobably, $\xi \sim \mathrm{Unif}[-.38, .38]$, and set

$$\ell(x; a, \xi) = x^2/2 - a\Psi(x) - \xi x, \qquad \Psi(x) = \begin{cases} x - x|x|, & |x| \le 1, \\ -x + \mathrm{sign}(x), & |x| > 1. \end{cases} \tag{16}$$

These coercive losses are C^1 with Lipschitz gradients and finitely many Hessian kinks. Their population loss is nonconvex.

Theorem 5 (Stationarity with positive occupied growth)**.** *At $\eta = 1$, SGD on ([16]) has a unique attracting invariant law supported on $[-1, 1]$, all moments finite, and stationary common-noise tangent exponent*

$$\lambda_\pi = \tfrac{1}{2}\log(1.16 \cdot 1.24) = 0.18176569\ldots > 0. \tag{17}$$

For the same loss, all $\eta \in [.98, 1]$ have a unique attracting invariant law and $\lambda_\pi \ge \log(1.1168) > 0$.

At $\eta = 1$, the update is the randomly perturbed tent map $X^+ = a(1-2\min\{|X|, 1\}) + \xi$. A two-step minorization gives uniform mixing, while its derivative magnitude on the occupied interval is exactly $2a > 1$, also giving an occupied harmonic gain above one. Appendix D proves both assertions and the step-size interval. This is an explicit SGD realization of a familiar expanding-map mechanism. It shows that attraction of *distributions* and growth of infinitesimal common-noise perturbations can coexist; no mode-count claim follows from the positive exponent.

5 From the theory to measurable neural quantities

The scalar reduction is exact in one dimension. In a network, it is a diagnostic of a specified frame. At state θ, let $J_B = I - \eta H_B(\theta)$ be the plain-SGD Jacobian and let q be a unit vector. Its scalar compression and reset norm are

$$a_q = q^\top J_B q, \qquad \|J_B q\|^2 = a_q^2 + \|(I - qq^\top)J_B q\|^2. \tag{18}$$

The second term is the motion discarded by scalar projection. Reinitializing q after every step also discards the direction inherited from earlier steps. The matrix-product measurement instead uses

$$v_{t+1} = J_{B_t}(\theta_t)v_t, \qquad \widehat{\lambda}_{[s,s+T)} = \frac{1}{T}\sum_{t=s}^{s+T-1} \log \frac{\|J_{B_t}(\theta_t)v_t\|}{\|v_t\|}. \tag{19}$$

Numerical normalization preserves v_t's direction and accumulates the logarithms. This is a finite-time transported growth rate, becoming a Lyapunov exponent only under a justified long-time limit. Even one-step norms do not determine this limit: alternating $\mathrm{diag}(2, .1)$ and $\mathrm{diag}(.1, 2)$ have norm two at each step but two-step product $.2I$.

For the batch-gradient axis $q_B = g_B/\|g_B\|$, with $g_B = \nabla L_B(\theta)$, we write $a_g = 1 - \eta s_B$, where $s_B = q_B^\top H_B q_B$. This uses the same batch for curvature and direction. We report $\widehat{\Lambda}_g = \mathrm{mean}\log|a_g|$ and $\widehat{H}_g = (\mathrm{mean}\,|a_g|^{-1})^{-1}$. These moments of a random compression do not equal moments of a batch-averaged sharpness, and q_B is neither fixed nor transported. A finite harmonic estimate also cannot certify that its population inverse moment is finite: rare near-zero gains can dominate it.

Finally, a projected parameter histogram must use one fixed axis and one fixed center. Otherwise the coordinate itself changes with the optimizer. We therefore pair dense fixed-frame traces with independent conditional probes and the derivative product in (19). Small windows reveal transient structure; bandwidth changes, ordered traces, and independent batch streams help distinguish a recurring pair of lobes from drift or a fragile histogram split.

6 Neural observations across regimes and windows

Neural diagnostics separate the two scalar edges. We study a three-convolution SiLU CNN and a two-hidden-layer, width-512 SiLU MLP, trained by plain SGD on the same 8,192-example CIFAR-10 subset with squared loss. At a fixed state, write $g_B = \nabla L_B(\theta)$, $s_B = g_B^\top \nabla^2 L_B(\theta) g_B/\|g_B\|^2$, and $a_g = 1 - \eta s_B$. Figure 2 estimates the log and negative-moment diagnostics of this scalar compression across batch sizes and step sizes. For the CNN, $(\eta, B) = (0.16, 128)$ gives $(\widehat{\Lambda}_g, \widehat{H}_g) = (0.179, 0.760)$, whereas $(0.08, 512)$ gives $(0.176, 1.176)$: similar positive log gain can occur on either side of the harmonic reference. Other settings approach one or both references; for example the MLP at $(0.02, 4096)$ has $(0.0027, 1.0008)$. The settings and checkpoint windows matter: the sweep averages do not assign an architecture a universal stability threshold.

A fixed coordinate reveals short-window structure. We continue a retained MLP checkpoint for 4,096 updates at $B = 4096$, $\eta = 0.04$. Its initial 128-draw conditional probe has $(\widehat{\Lambda}_g, \widehat{H}_g) = (0.136, 1.144)$; this selected checkpoint differs from the sweep's window averages. We freeze the reference leading-Hessian Ritz vector $q_\star$ and record $q_\star^\top(\theta_t - \theta_\star)$ before every update. The iid-minibatch control retains two-lobed occupancy (Figure 3); the displayed 512-update interval has lag-one correlation -0.985, consistent with approximate period-two alternation. Meanwhile, continuously propagated tangent vectors grow at approximately 0.00642 per update over the last 2,048 updates, while training loss decreases. Fixed-axis compression has a different late log average,

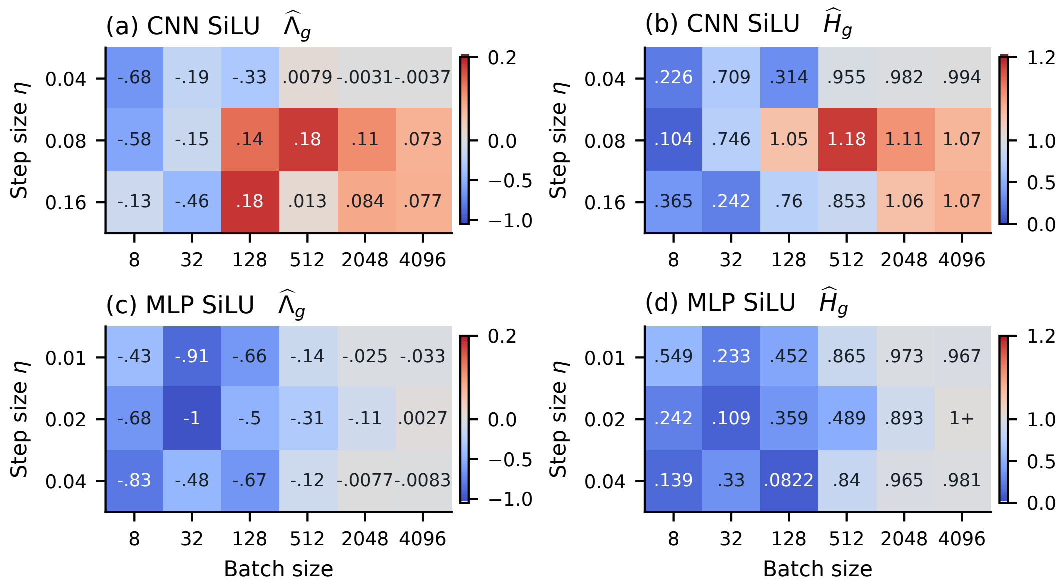

Figure 2: **Neural networks show distinct scalar diagnostic regimes.** Plain SGD on 8,192 CIFAR-10 examples with squared loss. For each run, the last three available conditional probe sets are pooled to estimate $\widehat{\Lambda}_g = \text{mean} \log |a_g|$ and $\widehat{H}_g = (\text{mean}\, |a_g|^{-1})^{-1}$; cells average these run-level statistics. Color centers are 0 and 1; axes use the actual training step size and batch size. CNN settings include $\widehat{\Lambda}_g > 0$ on both sides of $\widehat{H}_g = 1$. These are checkpoint-window diagnostics, with varying horizons, not equilibrium or full-network stability certificates. A trailing $+$ preserves a value just above the displayed reference after rounding.

-0.268, because it discards coupling to other directions. Adjacent occupied states can also lie on opposite sides of $\widehat{H}_{q_\star} = 1$. Thus the observations connect local expansion to bounded finite-time occupancy, but do not establish stationary bimodality, an asymptotic Lyapunov exponent, or a harmonic-threshold cause of neural bimodality. The CNN's discovery-selected 64-update candidate does not survive the same bandwidth screen in a separate continuation; the complete scans and this negative control are retained in the appendix.

Reading an architecture's operating regime. The two diagnostics in a cell use the same gain pool and scalar definition; comparisons across cells also change the trained states and checkpoint windows. The MLP at $\eta = .02$, $B = 4096$ approaches both references, whereas the CNN at $\eta = .16$, $B = 128$ is logarithmically expanding with H distinctly below one. The appendix adds ten further vision architectures: some approach the logarithmic reference, some the harmonic reference, and several remain contracting throughout their measured grid. This is a distribution of observed regimes across settings; there is no evidence here that architecture alone determines one preferred edge.

The chosen MLP checkpoint makes an additional distinction especially clear. Its initial batch-gradient law lies beyond both references, but its late fixed-leading-axis log average is negative even while transported tangents grow. The fixed axis retains the geometric identity of the two lobes; the tangent calculation retains the transported direction and its coupling to the other coordinates. Each is useful for a different question. The adjacent-state probes show that even within one oscillation, the fixed-axis harmonic statistic changes substantially. Averaging over the two phases can hide this structure.

Coverage and negative controls. The two MLP samplers both pass the fixed-top-axis shape screen in all windows of lengths 256, 512, and 1,024 in the declared scan. These overlapping windows document persistence across widths, not independent replications. The CNN supplies a useful contrast: its 64-update discovery interval fails the same screen in the confirmation stream, and neither continuation has a passing fixed-top-axis window of length 256 or more. Longer ordinary-network controls in the separate nonconvex campaign also show cutoff escape or parameter drift. Thus declining loss and a positive frozen diagnostic alone do not certify a neural equilibrium.

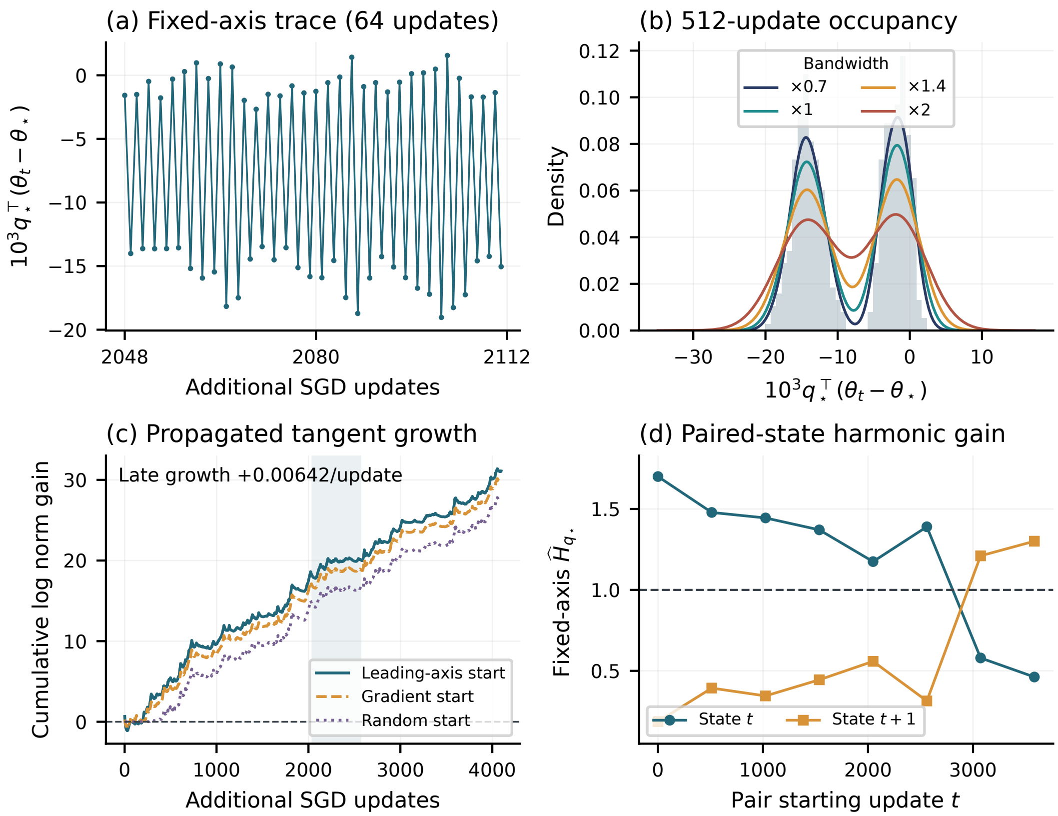

Figure 3: **Local expansion, two-lobed occupancy, and changing phase coexist.** SiLU MLP, CIFAR-10, $B = 4096$, $\eta = 0.04$, with independently sampled minibatches. The reference checkpoint $\theta_\star$ and its leading Hessian Ritz direction $q_\star$ are fixed; this coordinate origin need not be a density center. (a,b) A 64-update trace and its enclosing 512-update occupancy window show alternating lobes across four KDE bandwidths. (c) Continuously transported tangent vectors have late growth $+0.00642$ per update; loss decreases from 0.02485 to 0.00935. (d) The same 24 fresh minibatches probe each adjacent state pair; their fixed-axis harmonic gains lie on opposite sides of one. This is finite-window, approximately period-two occupancy; it does not establish a stationary bimodal law or a universal harmonic transition.

The supplementary atlas preserves missing probe cells and recorded cutoffs. It also includes a 168M-parameter GPT on FineWeb-Edu with token-aligned checkpoint pools. That model uses Adam, so its raw and frozen-preconditioner curvature views are explicitly auxiliary diagnostics; they do not measure the full optimizer-state Jacobian or inherit the plain-SGD theorems.

7 Relation to prior work and scope

The edge-of-stability observations of Cohen et al. (2021) motivate studying dynamics near the deterministic quadratic boundary. Batch sharpness and dynamical stochastic stability address different extensions of that picture (Andreyev & Beneventano, 2025; Chemnitz & Engel, 2025). Local quadratic models can remain predictive over substantial LLM training windows (Meterez et al., 2026); our distinction concerns what their scalar compressions imply about nonlinear return and occupation shape. Approximate period-two neural traces are also naturally related to two-step dynamics (Chen & Bruna, 2023).

Our probability tools have substantial classical foundations. Kesten–Goldie and matrix spectral renewal theory concern random recursions (Kesten, 1973; Goldie, 1991; Guivarc'h & Le Page, 2016); reflected-walk renewal supplies the scalar central exponent, and Markov renewal supplies its directional counterpart (Alsmeyer, 1997). Nonlinear asymptotically linear systems already have extensive stationary-tail theory (Alsmeyer, 2016; Alsmeyer et al., 2023). SGD heavy-tail analyses likewise precede this work (Gürbüzbalaban et al., 2021; Damek & Mentemeier, 2026). We do not claim a new general heavy-tail theorem or first discovery of confined chaotic optimization, which has deterministic and stochastic precedents (Chen et al., 2023; Tuci et al., 2026; Kalle & Maggioni, 2022).

The central renewal statements here have explicit entry and kernel assumptions. They are proved for the stated return models; their nonlinear transfer conditions have not been verified for ordinary neural SGD. Fixed additive Gaussian noise smooths the center, and matrix radial depletion need not produce two fixed-coordinate modes. The neural examples establish finite-window coexistence and diagnostic disagreement. They do not establish global parameter stationarity, a universal architecture-specific edge, or harmonic expansion as a cause of bimodality.

8 Discussion

Nonlinear return answers why local expansion need not cause escape. Negative moments answer a different question: under a controlled return law, how much stationary mass reaches the center? Matrix direction dynamics then determine which central law is relevant. The experiments show why all three questions matter: a network can lower its loss, display alternating two-lobe occupancy, and have different signs for fixed-axis and transported growth. A useful stochastic-edge description should therefore name its reference state, direction process, and time scale. Establishing a neural equilibrium law and its mode transition requires those measurements together with a verified return mechanism.